

Ques let DES be a block cipher where encryption of plaintext x with key k is denoted by -
 $y = \text{DES}(x, k)$

let $C(\cdot)$ represents bitwise complement operation defined by -

$$y' = \text{DES}(C(x), C(k))$$

Prove that $y' = C(y)$

i.e. show that complementing both the plaintext and key result in the complement of the ciphertext

Solⁿ

$$\text{let } y = \text{DES}(x, k)$$

$$y' = \text{DES}(C(x), C(k))$$

To prove $y' = C(y)$

DES working

1. XOR

2. Permutation

3. Sub keys

x	y	x'	y'	$x \oplus y$	$(x \oplus y)'$
0	0	1	1	0	1
0	1	1	0	1	0
1	0	0	1	1	0
1	1	0	0	0	1

$$(x \oplus y)' = x \oplus y' \text{ or } y \oplus x'$$

Permutation preserve complementation

$$\boxed{C(P(x)) = P(C(x)) \checkmark}$$

$$K_i' = C(K_i) \checkmark$$

$$L_i' = R_{i-1}$$

$$R_i' = L_{i-1} \oplus F(R_{i-1}, K_i)$$

$$L_i' = C(R_{i-1})$$

$$R_{i-1}' = C(R_{i-1})$$

$$K_i' = C(K_i)$$

$$L_i' = R_{i-1}' = C(R_{i-1})$$

$$R_i' = C(L_{i-1}') \oplus F(R_{i-1}', K_i') = C(L_{i-1}') \oplus C(F(R_{i-1}, K_i))$$

$$\overset{\text{sup}}{F}(C(R), C(K)) = C(F(R, K))$$

$$R_i' = C(L_{i-1}') \oplus C(F(R_{i-1}, K_i))$$

using XOR prop.

$$R_i' = C(L_{i-1}' \oplus F(R_{i-1}, K_i))$$

$$= C(R_i) \checkmark$$

$$(L_{i-1}', R_{i-1}') = (C(L_{i-1}), C(R_{i-1}))$$

$$L_i' R_i' = C(L_i), C(R_i)$$

$$\boxed{y' = C(y)}$$

Ques Consider the DES algorithm. Suppose the round function $F(R, K)$ is modified such that it ignores the key K and behaves in one of the following ways:

(a) $F(R, K) = 1$ (a 32-bit string of all zeros)

(b) $F(R, K) = R'$

For each case, answer the following

1. what overall function does DES compute after 16 rounds of encryption.

Solⁿ (i) $L_i = R_{i-1}$

$$R_i = L_{i-1} \oplus \underbrace{F(R_{i-1}, K_i)}_1$$

$$R_i = L_{i-1}$$

$$(L_1, R_1) = (R_0, L_0')$$

$$(L_2, R_2) = (L_0', R_0')$$

$$(L_3, R_3) = (R_0', L_0)$$

$$(L_4, R_4) = (L_0, R_0)$$

After every 4 round cycle will repeat it means after 16 round

$$\boxed{L_{16}, R_{16} = L_0, R_0} = \text{IP}^{-1} \text{ encr}$$

(ii)

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus \underbrace{F(R_{i-1}, K_i)}_{R_{i-1}}$$

$$R_i = L_{i-1} \oplus R_{i-1}$$

$$\overline{x \oplus y} = x \oplus \bar{y}$$

$$\overline{(x \oplus \bar{y})} = x \oplus y$$

$$(L_1, R_1) = (R_0, L_0 \oplus R_0')$$

$$(L_2, R_2) = (L_0 \oplus R_0', R_0 \oplus (L_0 \oplus R_0'))'$$

$$= (L_0 \oplus R_0', \bar{R_0} \oplus \bar{L_0} \oplus \bar{R_0})$$

$$= (L_0 \oplus R_0', L_0)$$

$$(L_3, R_3) = (L_0, L_0 \oplus R_0' \oplus L_0')$$

$$= (L_0, 1 \oplus R_0') = (L_0, R_0)$$

cycle length 3

$$16 \bmod 3$$

$$\boxed{1}$$

$$(L_{16}, R_{16})$$

$$= (R_0, L_0 \oplus \bar{R_0})$$